Unit 4

Measurement of Displacement and Strain

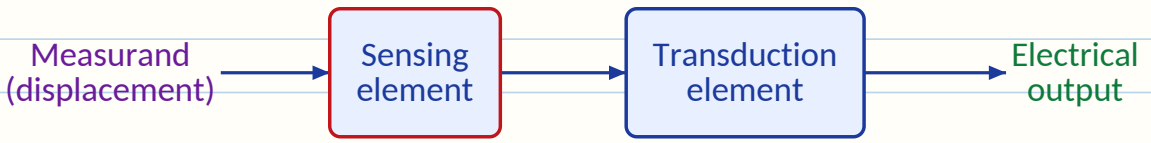
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Syllabus Topics Covered

1. Transducers: classification and requirements
2. Resistive (potentiometer) and inductive (LVDT) transducers
3. Capacitive transducers
4. Piezo-electric transducers
5. Ionization and photoelectric transducers
6. Calibration procedures
7. Electrical strain gauges and gauge factor
8. Use of strain gauges for bending, compression, tension and torque
9. Strain gauge rosettes

1. Transducers for Displacement Measurement

A **transducer** is a device that **converts one form of energy (the measured quantity) into another** - usually a **proportional electrical signal** (voltage, current, resistance, capacitance, inductance). **Displacement** (linear/angular position, thickness, level, strain, force, pressure, acceleration) is one of the commonly measured quantities.



Classification of transducers

Basis	Types	Examples

Active vs passive	**Active (self-generating)** : produce voltage/current without external power. **Passive** : need an **external excitation**	piezo-electric, thermocouple, photovoltaic / potentiometer, LVDT, strain gauge, capacitive
Primary vs secondary	primary senses (e.g. bellows -> displacement); secondary converts to electrical	diaphragm + LVDT
Analog vs digital	continuous output / pulse or coded output	LVDT / optical encoder
Principle	resistive, inductive, capacitive, piezo-electric, photoelectric, Hall effect, ionization	-

- **Requirements of a good transducer** : high **input-output linearity** , **sensitivity** , **repeatability** , **low hysteresis** , **good frequency response** , **small size / low weight** , minimum **loading effect** , **ruggedness** , **no sensitivity to environment (temperature, shock)** , **low cost** .

2. Resistive Transducers - Potentiometer

A **resistive potentiometer** (**pot**) has a **resistance element** (wire wound / conductive plastic / carbon film) with a **sliding contact (wiper)** attached to the moving object. The **output voltage** is a **fraction of the supply voltage** according to the **wiper position** .

Potentiometer

$e_o / e_i = x / x_t$ -> $e_o = e_i (x / x_t)$

- x = displacement of the wiper, x_t = total length of travel; e_i = excitation voltage (supply).
- Loading by a **load** R_m causes **non-linearity** ; the **loading error** is minimal when $R_m >> R_p$ (pot resistance).

- **Types** : **linear** (translatory, 1 mm to 500 mm), **rotary** (angular 0-360 degrees), **helical (multi-turn)** for **high resolution** . **Resolution** (for wire wound) = 1/N (N = no. of turns), about **0.1%** .
- **Advantages** : **simple, cheap, high output** , no amplifier required, both **AC & DC**

supply. **Disadvantages** : **wear** of contact (limited life), **friction** , **noise** , **low frequency response** , **resolution limited** , **loading effect** .

- **Uses** : **position feedback in servo mechanisms** , **machine tool slide position** , throttle/valve position, **level** (float) and **pressure** (Bourdon) indicators.

3. Inductive Transducers

Based on the **change in inductance** (self or mutual) of a coil due to the **movement** of a **ferromagnetic core / armature** , **change in air gap** or **coupling** . **Inductance L = N^2 / R_m** (N = turns, R_m = **reluctance** of the magnetic circuit).

3.1 Linear Variable Differential Transformer (LVDT)

The **most widely used inductive displacement transducer** : a **differential transformer** with **one primary and two identical secondary windings** , wound on a **hollow cylindrical former** ; a **movable soft-iron core** (ferromagnetic armature, moved by the **object** displacement) slides inside.

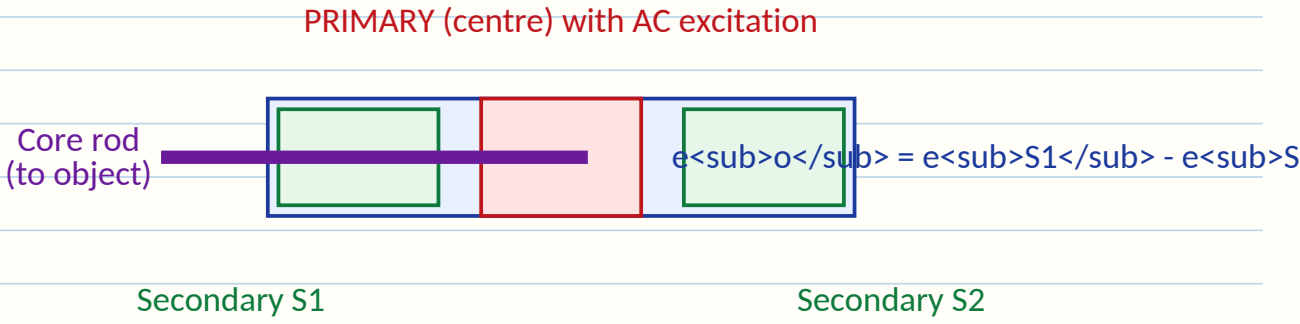


Fig 4.1: LVDT - construction

- **Working** : **AC (1-10 kHz, 3-15 V)** is applied to the primary. The **flux** links the secondaries through the **core** . **Secondaries are connected in series opposition** -> **net output e_o = e_S1 - e_S2** .

Core position	e_S1 vs e_S2	Output e_o
Central (null)	equal	**zero** (null position)
Displaced towards S1	e_S1 > e_S2	**in phase** with primary voltage; magnitude ∝ displacement
Displaced towards S2	e_S2 > e_S1	**180 degree out of phase** ; magnitude ∝ displacement

- **Phase-sensitive demodulator + filter** -> **DC output** with **polarity** indicating the **direction** . Output is **linear** over about **+-70% of the core range** .

- **Specifications** : range **+-0.1 mm to +-250 mm** ; **linearity 0.25%** ; sensitivity ~ 1-5 mV/V/mm; **resolution infinite** (practically 0.1 micron).
- **Advantages** : **no friction / wear** (no contact), **infinite resolution** , **high sensitivity** , **ruggedness** , **low hysteresis** , **repeatability** , **bidirectional** , **works in harsh environments** , **low power** consumption, **null reading is stable** . **Disadvantages** : need **AC excitation & demodulator** , **large core displacement** for small output, **sensitive to stray magnetic fields** , **dynamic response limited by carrier frequency (1/10)** .
- **Uses** : displacement (**gauging heads** , **comparators**), **thickness** , **pressure** (diaphragm), **force** (load cell), **acceleration** , **level** , **machine tool** position, **valve** position, **creep** and **roundness** .

3.2 Other inductive transducers

- **Variable reluctance** : **coil** + **moving armature** changes **air gap** -> inductance; **single (non-linear)** or **push-pull** (differential) types. Proximity sensing, **tachometer pick-up** (speed).
- **Mutual-inductance transducer** : two coils - mutual inductance changes with **separation / angle** (**synchros** , **resolvers** for angular position).
- **Eddy-current (proximity) transducer** : **high-frequency coil** - **eddy currents** in a nearby **conductive target** change the coil inductance / Q; measures **small distance (0.1 - 10 mm) non-contact** ; used for **shaft vibration** , **crack detection** , **thickness of coatings** .

4. Capacitive Transducers

Capacitance of a **parallel-plate capacitor** : **$C = \varepsilon_0 \varepsilon_r A / d$** ($\varepsilon_0$ = 8.854 pF/m; A = overlapping area; d = **gap** ; ε_r = dielectric constant). Displacement changes **d** , **A** or **ε_r** - hence **C** .



(a) change of gap d
C ∝ 1/d (non-linear)

(b) change of area A
C ∝ A (linear)

(c) change of dielectric
(level, thickness)

Principle	Relation	Remarks

Change of **gap** d	C = ε A / d; **ΔC / C = - Δd / d**	high sensitivity for **small displacements (um - mm)** ; **non-linear** -> use **differential** (push-pull) form
Change of **area** A	C ∝ A (linear)	**rotating vane** (angular) or **sliding plate** (linear); long range
Change of **dielectric** ε_r	slab of dielectric moves in/out	**liquid level** , **thickness / moisture** measurement

- **Measuring circuits** : **AC bridge** , **oscillator (frequency-modulated)** , **charge/voltage amplifier** , **operational amplifier** (output ∝ d for gap type: $e_o = - e_i (C_i / C_x) \propto d$).
- **Advantages** : **very small force needed** , **high sensitivity** , **excellent frequency response (MHz)** , **non-contact** , works at **high temperature** , **resolution < 0.01 micron** . **Disadvantages** : **stray capacitance** and **cable effects** , **temperature / humidity** , **non-linear** (gap type), needs **complex electronics** , **insulation** problems.
- **Uses** : **gap / thickness** (rolled sheets), **vibration** , **proximity** switches, **touch screens** , **liquid level** , **pressure (diaphragm)** , **micrometric positioning** (semiconductor lithography).

5. Piezo-electric Transducers

Piezo-electric effect : certain **crystals** (**quartz** , Rochelle salt, barium titanate, lead zirconate titanate **PZT**) develop an **electric charge** on their surface when **mechanically stressed** (force). The **reverse** effect: **deform** under an electric field (**ultrasonic generators**). Self-generating (**active**) transducers.

Charge and voltage

Q = d F (charge Q, force F, piezo-electric constant d (C/N): quartz **d = 2.3 pC/N**)

E_o = Q / C_p = d F / C_p , or **E_o = g t P** (g = voltage sensitivity (V m/N), t = thickness, P = pressure)

- C_p = capacitance of the crystal + cable.

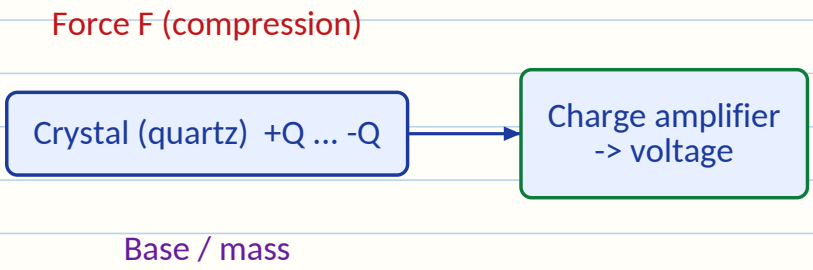


Fig 4.2: Piezo-electric transducer

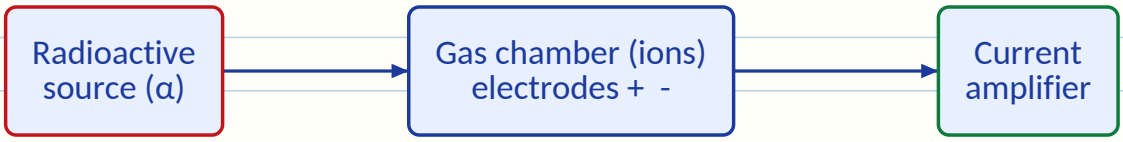
- **Measurement of displacement** : the force from a **spring-mass** (seismic) system or **diaphragm** is applied to the crystal; the **charge / voltage** is proportional to **force -> displacement, pressure, acceleration** . Practical for **dynamic** (rapidly changing) displacement / vibration.
- **Charge amplifier** (high input impedance): converts charge to voltage and **removes cable capacitance effect** ; **low-noise cable** needed.
- **Advantages** : **very high frequency response (up to 100 kHz+)** , **small, rugged** , **self-generating** , **high output** , wide range, **insensitive to temperature (quartz)** , **no external power** . **Disadvantages** : **can't measure static** quantities (charge **leaks** away: **only dynamic**), **high-impedance** electronics, **temperature effects (ceramics)** , **cable noise** , **pyro-electric** effects.
- **Uses** : **accelerometers** , **pressure** (engine knock, shock), **force / load cells** (dynamic), **vibration** , **ultrasonic transducers** , **igniters** , **microphones** , **acoustic emission** , **quartz crystal oscillator (clocks)** .

6. Ionization & Photoelectric Transducers

6.1 Ionization transducer

Ionization transducers use the **ionization of a gas** by a **radioactive source** (alpha particles) or by **an electric discharge** ; the **ion current** between two electrodes changes with the **geometry** (distance between electrodes, gas pressure).

- **Principle** : A **small radioactive source (Radium-226 / Americium-241 / Polonium)** ionizes the gas between **two electrodes** (**anode and cathode**) kept at a **potential** ; **ion current** (**saturation**) ∝ **number of ions collected** , which depends on the **path** and **gas pressure/density** . If one electrode **moves** (displacement), the **ion current changes** -> measure **displacement** with an **electrometer / amplifier** .



- **Other forms** : **ionization gauge (Bayard-Alpert)** for **low pressure (vacuum)** measurement; **flame ionization** & **smoke detectors** ; **radiation (beta gauge) thickness gauges** where absorption of rays through the **material** is measured.
- **Advantages** : **non-contact** , **high sensitivity** to small displacements, **no moving parts** , **very high-temperature** possible. **Disadvantages** : **radiation hazard** , **expensive** , **shielding** and **licence** , temperature/pressure dependence, **limited use** in practice.

6.2 Photoelectric transducers

Convert **light energy -> electrical signal** ; used for **non-contact** displacement, **position** , **counting** , **speed** and **dimension** measurement.

Type	Principle	Examples
Photo-emissive	light **ejects electrons** from a **cathode** in a vacuum tube -> **photo-current** ∝ intensity	phototube, **photomultiplier**
Photo-conductive	**resistance falls** when light falls on **semiconductor**	**LDR (CdS)** , photoconductive cell
Photo-voltaic	**generates EMF** when light falls on a **p-n junction**	**solar cell** , **photodiode** , **phototransistor**

- **Displacement measurement** : a **moving shutter / slit / mask** (attached to the object) **varies the amount of light** reaching a **photocell** -> output ∝ displacement (**linear**). **Reflective type** : **light reflected** from the object surface changes with the distance (**fibre-optic displacement sensor**).
- **Optical encoders** : **disc with transparent / opaque slots** (**incremental** pulses; **absolute** binary/gray code) with **LED + photodetector** -> **digital output** of angular / linear position (**resolution 1 / 10^4 turn** , **linear glass scales 0.5 - 5 micron**); also **laser (interferometer)** displacement.
- **Advantages** : **non-contact** , **fast response** , **high resolution** , **digital output** , **no wear** . **Disadvantages** : **ambient light** , **dust / oil contamination** , **temperature drift** , **aging** of cells, alignment of **optics** .

7. Calibration of Displacement Transducers

Calibration establishes the **relationship between the transducer output and the actual displacement** by comparing with a **standard** of higher accuracy.

1. Mount the transducer **rigidly** on a **calibrating stand** (**slip gauges** , **micrometer head** , **precision slide with vernier** or **laser interferometer** as reference).
2. **Warm up** the instrument; **adjust the zero** (null) and the **gain (span)** with the lowest & highest standard displacements.
3. Apply **known displacements** (e.g. 0, 10, 20, ..., 100% of range) in **steps upward (ascending)** and then **downward (descending)** , **recording the output** at each step (**3 to 5 cycles**).
4. **Plot** the **calibration curve** (output vs displacement): find **sensitivity (slope)** , **linearity** (best-fit straight line, % of full scale), **hysteresis** (difference up/down), **repeatability** , **zero drift** , **resolution** .
5. **Prepare a calibration chart / correction table** ; **certify** ; label the **next due date** .

- **Dynamic calibration** : apply a **step or sinusoidal** motion by a **vibration exciter (shaker)** to find **frequency response** and **time constants** .

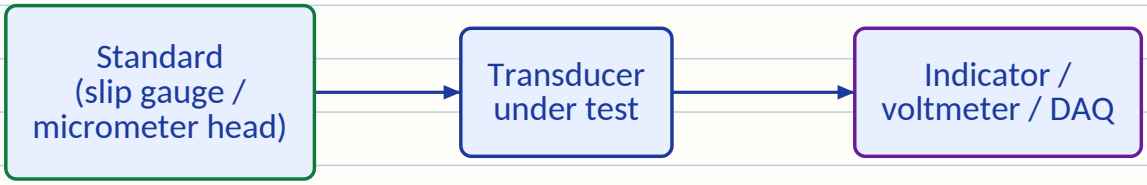


Fig 4.3: Static calibration set-up

8. Measurement of Strain

Strain (ε) = change in length / original length = ΔL / L (dimensionless; micro-strain, με = 10^{-6}). Under **Hooke's law σ = E ε** ; hence **stress can be found from strain** . **Electrical strain gauges** are the **most widely used** : they convert **strain into a change in resistance** .

Strain gauge principle & gauge factor

The **resistance of a conductor** R = ρ L / A. When the conductor is **stretched** , **length increases** , **area decreases** and **resistivity ρ changes** (piezo-resistive effect):

Gauge factor (strain sensitivity)

GF = (ΔR / R) / ε = 1 + 2ν + (Δρ/ρ)/ε

- ν = **Poisson's ratio** (0.3 for most metals) -> **1 + 2ν ≈ 1.6** ; the **piezo-resistive term** adds ~ 0.4 -> **GF ≈ 2 (metal foil: 1.9-2.1)** .
- **Semiconductor gauges (Si)** : **GF = 100 - 200** (up to +- 175; large piezo-resistive contribution).
- **ΔR = GF x R x ε** , e.g. R = 120 ohm, GF = 2, ε = 1000 με -> ΔR = 0.24 ohm.

Types of electrical strain gauges

Type	Construction	Features
Unbonded wire	wire stretched over insulated **pins** between a fixed frame and a movable armature	used in **transducers** (force, acceleration, pressure); **no cement** ; high stability
Bonded wire (grid)	**fine wire (0.025 mm)** in **grid** on **paper / plastic backing** , cemented to specimen	cheap; **earliest** type; poor heat dissipation
Metal foil	**etched foil grid** (2-5 micron) on **polyimide / epoxy backing** ; **most common**	**better heat dissipation** , **flexible shape** , **good fatigue life** , **stable**
Semiconductor	**silicon** strip with leads; **piezo-resistive**	**high GF 100-200** ; **non-linear** , temperature-sensitive, **brittle**
Weldable / capacitive / thin film	**spot welded** ; **capacitive** for high temp; **sputtered** film on diaphragm	**high-temperature** , **long term** field use

- **Materials** : **Constantan (Cu-Ni 45-55)** (GF 2.1, low temp coefficient - **most common**), **Isoelastic (Fe-Ni-Cr-Mo)** (GF 3.6, **dynamic**), **Nichrome V** (GF 2.1), **Karma** , **Platinum-tungsten** (GF 4, **high temp**), **Silicon** .
- **Gauge length** 0.2 - 100 mm (standard 6 mm); **resistance 120, 350, 1000 ohm** ; **backing** : paper, **polyimide** , **glass-fibre-reinforced epoxy** ; **adhesive** : cyanoacrylate, epoxy (**strain-transfer layer**).

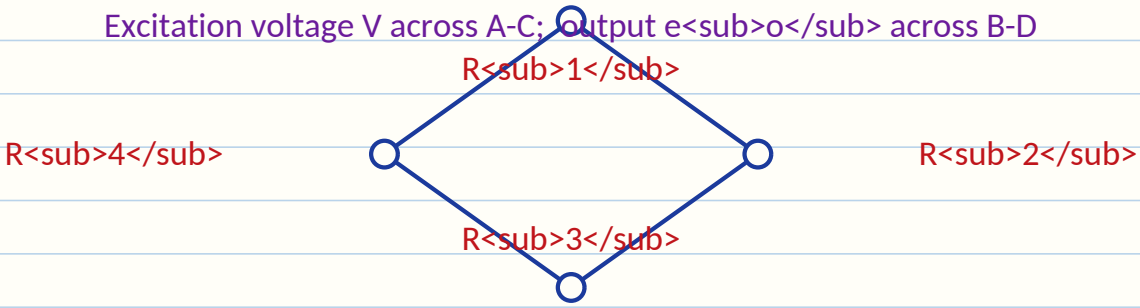
Installation (bonding) procedure

1. **Surface preparation** : degrease, abrade (emery 220-320 grit), **clean** with solvent (acetone), **mark alignment lines** , neutralize.
2. **Position** gauge on the marked axis, **apply adhesive** (thin), press with **thumb**

(gauge pressure) and **cure** (time/temperature).

3. **Solder leads** (**terminals**), **protective coating** (**moisture-proof** : wax, silicone), **check resistance & insulation** (> 1000 M-ohm).

Measuring circuit: Wheatstone bridge



Bridge output

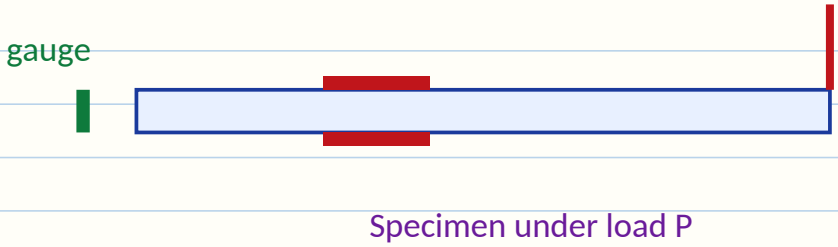
$e_o = V [R_1 / (R_1 + R_2) - R_4 / (R_3 + R_4)]$; balanced when $R_1R_3 = R_2R_4$.

- **Quarter bridge** (one active gauge): $e_o = V (\Delta R/R)/4 = V\ GF\ \varepsilon / 4$.
- **Half bridge** (2 active gauges): $e_o = V\ GF\ \varepsilon / 2$ (e.g. bending).
- **Full bridge** (4 active): $e_o = V\ GF\ \varepsilon$ - **largest output, temperature compensated** .

- **Temperature compensation** : a **dummy gauge** (same material & type, unstrained, on the same material) in the **adjacent arm** of the bridge cancels the resistance change due to temperature; **self-temperature-compensated (STC)** gauges are also available.
- **Example** : V = 10 V, GF = 2, ε = 500 με, quarter bridge -> $e_o = 10 \times 2 \times 500 \times 10^{-6} / 4$ = **2.5 mV** ; **full bridge = 10 mV** .

9. Use of Strain Gauges - Bending, Tension, Compression, Torque

9.1 Tension and compression (axial loading)



Specimen under load P

- **Single active gauge** on the member along the load axis: $\varepsilon = \sigma / E = P / (A E)$;

$\Delta R/R = GF\ \varepsilon$; **load $P = \varepsilon A E$** .

- **Poisson gauge** (**second gauge transverse** , inactive for axial strain but $\varepsilon_t = -\nu\ \varepsilon$) in an **adjacent arm** -> **output increases by (1 + ν)** : $e_o = V\ GF\ \varepsilon\ (1 + \nu) / 4$. **Dummy gauge** gives **temperature compensation** .
- **Four-gauge full bridge** on a **column / tension bar** : two axial (R_1, R_3, opposite arms) + two transverse (R_2, R_4): **output = V GF ε (1 + ν) / 2** ; **eliminates bending effects & temperature** . Used for **load cells** .

9.2 Bending (cantilever / beam)

- In a **cantilever** beam loaded at the free end, the **top fibre is in tension (+ε)** and the **bottom fibre in compression (-ε)** with equal magnitude. Mount **gauge R_1 on top and R_2 on bottom** , in **adjacent arms** of the bridge -> **output doubles** (half bridge: $e_o = V\ GF\ \varepsilon / 2$), **temperature compensated automatically** , **axial load cancelled** .
- **Strain at gauge : $\varepsilon = 6 P L / (E b t^2)$** (rectangular section, L = distance from the load to the gauge, b = width, t = thickness); **bending moment M = P L** ; **stress $\sigma = 6M/(b t^2) = E \varepsilon$** .
- **Full bridge for bending : two gauges on top (R_1, R_3) + two on bottom (R_2, R_4)** -> output **= V GF ε** (4 times the quarter-bridge). Used in **bending-beam load cells** , **weighing scales** .

9.3 Torque measurement (shaft)

A shaft in **pure torsion** has **principal strains at +45 degrees (tension) and -45 degrees (compression)** to its axis; **shear strain γ** along the axes. **Gauges are bonded at 45 degrees** to the shaft axis.

Shaft with 4 gauges at +-45 degrees (full bridge)



Torque from strain

Shear stress $\tau = G\gamma$; $\varepsilon_{45} = \gamma / 2 = \tau / (2G)$ (magnitude equal at +45 and -45; opposite sign).

$\tau = G\gamma = 2G\varepsilon_{45}$; solid shaft: $T = \pi d^3 \tau / 16 = \pi d^3 G \varepsilon_{45} / 8$, and with $G = E / 2(1 + \nu)$: $T = \pi d^3 E \varepsilon_{45} / (16(1 + \nu))$.

- **Full bridge (4 gauges, +45, -45, +45, -45)** : $e_o = V\,GF\,\varepsilon_{45}$ - four times output, **cancels bending & axial loads** and temperature. **Slip rings / telemetry** needed for rotating shafts.

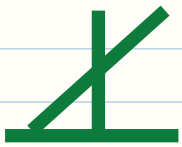
- **Example** : solid steel shaft d = 40 mm, E = 200 GPa, ν = 0.3, measured ε_{45} = 500 με -> $T = \pi \times 0.04^3 \times 200e9 \times 500e\text{-}6 / (16 \times 1.3) = 966$ N.m (approx.).

Practical points

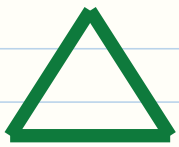
- **Use of dummy gauges** & **bridge balancing** ; **lead-wire resistance** compensation (**three-wire** connection); **shunt calibration** (a known resistor across one arm simulates a known strain); **zero drift** , **self-heating** (limit gauge current < 25-30 mA), **hysteresis** , **creep** , **cross sensitivity** (transverse strain).

10. Strain Gauge Rosettes

When the **directions of principal strains are unknown** (a biaxial stress field), strain at a point must be measured in **at least three directions** . A **rosette** = **three (or more) strain gauges arranged at fixed angles** on a **common backing** .



Rectangular rosette
0 / 45 / 90 degree



Delta rosette
0 / 60 / 120 degree



Tee rosette (two gauges at 90 degree)

Rectangular (0 - 45 - 90) rosette: gauges a, b, c with strains ε_a, ε_b, ε_c

Principal strains : $\varepsilon_1,2 = (\varepsilon_a + \varepsilon_c) / 2 \pm (1/\sqrt{2})\ \mathrm{sqrt}[\ (\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2\ ]$

Direction : $\tan 2\theta = (2\varepsilon_b - \varepsilon_a - \varepsilon_c) / (\varepsilon_a - \varepsilon_c)$ (θ from gauge a to ε_1)

Principal stresses : $\sigma_1 = E(\varepsilon_1 + \nu\varepsilon_2) / (1 - \nu^2)$, $\sigma_2 = E(\varepsilon_2 + \nu\varepsilon_1) / (1 - \nu^2)$

Delta (0 - 60 - 120) rosette

$\varepsilon_{1,2} = (\varepsilon_a + \varepsilon_b + \varepsilon_c) / 3 \pm (\sqrt{2} / 3) \sqrt{[(\varepsilon_a - \varepsilon_b)^2 + (\varepsilon_b - \varepsilon_c)^2 + (\varepsilon_c - \varepsilon_a)^2]}$

Example (rectangular rosette) : $\varepsilon_a = 800$, $\varepsilon_b = 400$, $\varepsilon_c = -200$ με -> $(\varepsilon_a + \varepsilon_c)/2 = 300$; $\sqrt{[(400)^2 + (600)^2]} = 721.1$; $/\sqrt{2} = 509.9$ -> **$\varepsilon_1 = 810$ με, $\varepsilon_2 = -210$ με** ; $\tan 2\theta = (800 - 800 + 200)/(1000) = 0.2$ -> **$\theta = 5.7$ degree** .

- **Stacked / planar rosettes** : **stacked** (gauges one above another) for **small spots** ; **planar** (side by side) for **cooler operation** . **Uses** : **stress analysis** of **pressure vessels, aircraft structure, bridges** , **welded joints** , **residual stress** (hole drilling).
- **Selection** : **rectangular** for **known principal directions** or **simple** reduction; **delta** has **equal angle** (uniform), **better** when **no preference** about direction.

Quick Revision / Important Questions

1. LVDT: construction, working, characteristics, advantages. Potentiometer and loading effect.
2. Capacitive and piezo-electric transducers; charge amplifier.
3. Ionization and photoelectric transducers; calibration procedure.
4. Strain gauge: gauge factor derivation, types, Wheatstone bridge, temperature compensation.
5. Strain gauges for bending, tension / compression and torque; rosettes (examples).